

DeepThinkTrader v1.0 – Complete System Guide

Generated: March 18, 2026 | **Repository:** github.com/RebelHawk-TK/DeepThinkTrader

CRITICAL WARNING: This is a paper trading bot. Never use real money until backtested for 3+ months. Trading bots can lose 100% of capital. This project is educational and experimental.

1. Architecture Overview

Pipeline: Scanner → Research Agent → AI DeepThink (Claude) → Execution Agent → Alpaca Paper Trading

Component	Technology	Purpose
Scanner Agent	Alpaca Screener API	Discovers 100 stocks/day from most active, top movers, news trending, popular large-caps
Research Agent	Alpaca + Twelve Data + NewsAPI + Yahoo Finance	Gathers technicals, fundamentals, news sentiment, analyst ratings, insider trades, earnings data
AI DeepThink Agent	Claude Haiku 4.5 (Anthropic API)	Chain-of-thought analysis using 5 institutional frameworks
Execution Agent	Alpaca Paper Trading API	Places trades with 8 risk guardrails, captures X-Request-ID
Dashboard	Streamlit + Plotly	Real-time portfolio monitoring, trade history, analyses
Database	SQLite	Trade logging, research reports, API request IDs

2. Data Sources

Source	Data Provided	API Key Required	Free Tier
Alpaca Markets	Price, volume, SMA, RSI, trade execution, portfolio history	Yes (paper account)	Unlimited
Twelve Data (RapidAPI)	MACD, Bollinger Bands, EMA 9/21, Stochastic, ADX, ATR	Yes (RapidAPI key)	800 req/day
NewsAPI	News headlines + VADER sentiment scoring	Yes	500 req/day

Yahoo Finance	P/E, revenue growth, margins, analyst targets, earnings dates, insider trades, institutional holdings	No	Unlimited
Anthropic Claude	AI chain-of-thought analysis	Yes	Pay per token (~\$0.20/cycle)
Reddit (PRAW)	Sentiment from r/wallstreetbets, r/stocks, r/investing	Yes (optional)	60 req/min

3. AI Analysis Frameworks

Every stock is analyzed by Claude through **5 institutional-grade frameworks**:

3.1 Citadel Technical Analysis (25%)

Signal	BUY	SELL
MACD	Bullish crossover + histogram positive	Bearish crossover + histogram negative
RSI	30-50 recovering from oversold	> 70 with bearish divergence
Stochastic	Bullish cross in oversold zone ($K < 20$)	Bearish cross in overbought zone ($K > 80$)
Bollinger	Bounce off lower band / squeeze breakout	Break below middle band, expanding bandwidth
ADX	> 25 confirming uptrend	> 25 confirming downtrend
Volume	> 1.5x average (accumulation)	Declining on advances

3.2 Goldman Sachs Fundamental Analysis (25%)

- Analyst consensus BUY with > 15% upside to target
- Revenue growth > 10% with improving margins
- P/E below sector average OR PEG < 1.5
- Insider net buying in last 30 days
- Earnings beat rate > 75% last 4 quarters

3.3 Catalyst Scoring (20%)

- Positive news (new product, partnership, upgrade) with impact > 5/10
- Social sentiment turning positive
- Sector tailwinds
- **No catalyst = no trade** (most important rule)

3.4 Renaissance Technologies Pattern Detection (15%)

Seasonal Patterns:

- Monday weakness, Friday drift up
- Month-end institutional rebalancing
- January small-cap effect
- "Sell in May" seasonal weakness

Institutional Signals:

- Ownership > 70% = follow smart money
- Increasing ownership = accumulation (bullish)

Insider Patterns:

- Cluster buying (3+ insiders) = very strong bullish
- CEO/CFO buying with personal money = strongest signal
- Sudden large sales outside pattern = red flag

Price Anomalies:

- Post-earnings drift (60 days)
- Mean reversion (> 2 std dev from 50-day)

- Decreasing ownership = distribution (bearish)
- Short interest > 15% = squeeze potential
- Volume spike + flat price = accumulation before move

3.5 Bridgewater Risk Management (15%)

Check	Rule	Action
Sector Concentration	Same sector > 30% of portfolio	Reduce size 50% or HOLD
Correlation	Highly correlated with existing position	Flag in risks, prefer uncorrelated
Stress Test	5% market drop would cause > 8% portfolio drawdown	HOLD — don't add risk
Liquidity	Avg volume < 1M shares/day	Flag as low liquidity
Single Position	Position > 10% of account	Reduce size
Beta Adjustment	Beta > 1.5	Reduce size 30%

4. JPMorgan Earnings Framework

Days to Earnings	Action
> 14 days	Trade normally
7-14 days	Reduce position size 30%
3-7 days	HOLD (unless explicit earnings play with conviction 9+)
< 3 days	AUTOMATIC HOLD — binary event risk

Earnings Play Criteria (ALL must be met, conviction 9+):

- Beat rate > 75% last 4 quarters
- Revenue growth accelerating QoQ
- Estimates recently revised upward
- Historical post-earnings reaction positive
- Options implied move < historical average (market underpricing)

5. Risk Guardrails (Execution Agent)

#	Rule	Action if Triggered
1	Conviction < 8/10	HOLD
2	Max loss per position > 2% of account	BLOCKED
3	Take-profit < 2x stop-loss	BLOCKED
4	Daily realized loss > 5% of account	BLOCKED for rest of day
5	Already 5 open positions	BLOCKED
6	Duplicate position in same ticker	BLOCKED
7	Outside US market hours	BLOCKED
8	3+ consecutive losses (revenge trading)	BLOCKED + cooldown

6. Configuration (.env)

```
# Alpaca API (Paper Trading)
ALPACA_API_KEY=your_key
ALPACA_SECRET_KEY=your_secret
```

```
ALPACA_BASE_URL=https://paper-api.alpaca.markets
```

```
# Anthropic (Claude API)
```

```
ANTHROPIC_API_KEY=your_key
```

```
# RapidAPI (Twelve Data)
```

```
RAPIDAPI_KEY=your_key
```

```
# NewsAPI
```

```
NEWSAPI_KEY=your_key
```

```
# Reddit (Optional)
```

```
REDDIT_CLIENT_ID=your_id
```

```
REDDIT_CLIENT_SECRET=your_secret
```

```
REDDIT_USER_AGENT=DeepThinkTrader/1.0
```

```
# Trading Parameters
```

```
ACCOUNT_SIZE=100000
```

```
WATCHLIST=NVDA,TSLA,AAPL,AMD,SPY
```

```
MAX_RISK_PER_TRADE=0.02
```

```
MAX_DAILY_LOSS=0.05
```

```
MIN_CONVICTION=8
```

```
RESEARCH_INTERVAL_MINUTES=60
```

```
MIN_REWARD_RISK_RATIO=2.0
```

7. Quick Start

```
git clone https://github.com/RebelHawk-TK/DeepThinkTrader.git
cd DeepThinkTrader
cp .env.template .env          # Fill in your API keys
pip install -r requirements.txt
python main.py once             # Test single cycle
python main.py scan             # Test scanner
streamlit run dashboard.py      # Launch dashboard
./run.sh                       # Start bot + dashboard
./stop.sh                      # Stop everything
./status.sh                    # Check status
./install.sh                   # Install as macOS background service
```

8. Estimated Costs

Service	Cost/Cycle (100 stocks)	Cost/Day (~7 cycles)	Cost/Month
Claude Haiku 4.5	~\$0.20	~\$1.40	~\$42
Alpaca	Free	Free	Free
Twelve Data	Free (800 req/day)	Free	Free
NewsAPI	Free (500 req/day)	Free	Free
Yahoo Finance	Free	Free	Free
Total			~\$42/month

9. Project Structure

```
StockTrader/
├─ main.py           # Orchestrator (scheduled loop)
├─ config.py         # Environment config
├─ dashboard.py      # Streamlit monitoring
├─ run.sh / stop.sh / status.sh # Control scripts
├─ install.sh        # macOS launchd installer
├─ agents/
│   ├─ ai_deepthink_agent.py # Claude AI analysis (5 frameworks)
│   ├─ deepthink_agent.py   # Rule-based fallback
│   ├─ execution_agent.py    # Alpaca trades + risk guardrails
│   ├─ research_agent.py     # Multi-source data gathering
│   └─ scanner_agent.py      # 100-stock daily discovery
```

```
|— utils/
|   |— alpaca_data.py      # Alpaca market data + X-Request-ID
|   |— database.py        # SQLite trade logging
|   |— risk_manager.py    # Position sizing + daily limits
|   |— twelve_data.py     # MACD, Bollinger, EMA, Stochastic, ADX, ATR
|   └─ yahoo_fundamentals.py # P/E, earnings, analysts, insiders
|— .env.template          # API key template
|— requirements.txt        # Python dependencies
└─ docs/                  # Documentation + setup guide PDF
```

DeepThinkTrader v1.0 — Paper Trading Mode Only — github.com/RebelHawk-TK/DeepThinkTrader

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